BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LANs, WANs and Network Models

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Convert 00101101 binary to denary.

2. What is the Internet?

A) A network of networks
B) One central web server
C) Only public web pages
D) One school's wireless LAN

3. "A web client must host every website before it can request a page." Correct this. Explain why.

4. A pupil needs work on several devices. "Cloud files are always available when the Internet fails." Correct this. What could happen?

Today's objectives

Explain the core ideas of lans, wans and network models using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

LAN
Covers a small geographical area such as one building, usually under a single organisation.

WAN
Connects networks over wider geographical areas, often via third-party infrastructure.

Client-server
Central servers provide resources/services for client devices.

Peer-to-peer
Devices share resources directly without a dedicated central server for those resources.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

LAN	Covers a small geographical area such as one building, usually under a single organisation.
WAN	Connects networks over wider geographical areas, often via third-party infrastructure.
Client-server	Central servers provide resources/services for client devices.
Peer-to-peer	Devices share resources directly without a dedicated central server for those resources.

I Do - One worked example

A company has two offices in different cities connected so staff share files. Identify network types.

1. Each office can use its own LAN.
2. A WAN connects geographically separate LANs.
3. A server can centrally control shared files and access.
4. Explain why peer-to-peer may be less suitable for many managed users.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Compare a five-computer home network with a national retailer's network.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the missing output for a connection between two distant school sites.

```
separateSites = true
if separateSites then
...
else
print("One LAN")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Explain one difference between LAN and WAN. [2 marks]

Q3. Give one advantage of client-server networking. [2 marks]

Q4. Explain one drawback of peer-to-peer for a large school. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A WAN can connect LANs in geographically separate locations. ☐ **T** ☐ **F**
2. Covers a small geographical area such as one building, usually under a single organisation. ☐ **T** ☐ **F**
3. Connects networks over wider geographical areas, often via third-party infrastructure. ☐ **T** ☐ **F**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 45, because 32 + 8 + 4 + 1 = 45. ☐

Q2 A - A network of networks ☐

Q3 A server stores and makes website resources available for clients to request. ☐

Q4 Cloud resources are accessed over a network. Files may be unavailable if the connection or server fails. ☐

Code First - Model

```
separateSites = true
if separateSites then
print("WAN connects sites")
else
print("One LAN")
endif
```

Output: WAN connects sites

We Do - Check the model

1. Home is LAN; retailer uses LANs connected by WAN. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 LAN covers smaller/local area; WAN connects over wide geographical areas. ☐

Q3 Centralised user account, access or resource management. ☐

Q4 Harder to centrally manage permissions, security or backups across many devices. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: